Appendix

Table A1. PRISMA checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction, Paragraphs 1-2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Introduction, Paragraph 3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Methods, Section 2.3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Methods, Section 2.2
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Methods, Section 2.3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Methods, Section 2.3
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Methods, Section 2.3
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Methods, Section 2.3
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Methods, Section 2.4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Tables 1-2
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Methods, Section 2.5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data	Methods, Section 2.5

Section and Topic	Item #	Checklist item	Location where item is reported
		conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Methods, Section 2.5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Methods, Section 2.5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Methods, Section 2.5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Methods, Section 2.5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Methods, Section 2.5
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Methods, Section 2.5
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix
Study characteristics	17	Cite each included study and present its characteristics.	Tables 1-2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Results, Section 3.5; Appendix
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Tables 1-2; Figures 2-3
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Results, Section 3.2-3.3; Appendix
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 2-3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Results, Section 3.2-3.3
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Results, Section 3.2-3.3
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Results, Section 3.2-3.3
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Results, Section 3.2-3.3
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Discussion
	23b	Discuss any limitations of the evidence included in the review.	Discussion, Section 4.1
	23c	Discuss any limitations of the review processes used.	Discussion,

Section and Topic	Item #	Checklist item	Location where item is reported
			Section 4.1
	23d	Discuss implications of the results for practice, policy, and future research.	Discussion; Conclusions
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Methods, Section 2.1
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Methods, Section 2.1
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding
Competing interests	26	Declare any competing interests of review authors.	Declaration of interest
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Manuscript; Appendix

Table A2. Search strategy.

#	Searches	Results
1	Perioperative Medicine/ [new MeSH 2020]	62
2	Anesthesia Department, Hospital/	1088
3	Anesthesiology/	31896
4	Anesthetists/	354
5	Anesthesiologists/	1298
6	Nurse Anesthetists/	2779
7	Perioperative care/	15033
8	exp Preoperative care/	70983
9	exp Preoperative Period/	8493
10	Perioperative Period/	3565
11	exp Perioperative Nursing/	13803
12	Perioperative Nursing/	6851
13	exp Professional-Patient Relations/ and anesthe*.mp,jw,kf.	1047
14	Ambulatory Care Facilities/ and (exp "Anesthesia and Analgesia"/ or an?esthe*.mp,jw,kw.)	283
15	(before adj2 operat????).mp,kw.	16122

16	(before adj2 surgery).mp,kw.	43123
17	(before adj3 procedur*).mp,kw.	9835
18	(prior adj2 operat????).mp,kw.	2332
19	(prior adj2 surgery).mp,kw.	15132
20	(prior adj3 procedur*).mp,kw.	3933
21	perian?esth*.mp,kw.	538
22	peri-an?esth*.mp,kw.	68
23	perioperat*.mp,kw.	100376
24	peri-operat*.mp,kw.	6603
25	periprocedur*.mp,kw.	5160
26	peri-procedur*.mp,kw.	1069
27	peroperat*.mp,kw.	4618
28	peroperat*.mp,kw.	4618
29	preoperat*.mp,kw.	304652
30	pre-operat*.mp,kw.	29786
31	preprocedur*.mp,kw.	3507
32	pre-procedur*.mp,kw.	2122
33	preadmission?.mp,kw.	1443
34	pre-admission?.mp,kw.	1002
35	preanesth*.mp,kw.	8869
36	pre-anesth*.mp,kw.	442
37	preanaesth*.mp,kw.	295
38	pre-anaesth*.mp,kw.	338
39	presurger*.mp,kw.	1260
40	pre-surger*.mp,kw.	1236
41	presurgical.mp,kw.	7852
42	pre-surgical.mp,kw.	2500
43	(ahead adj3 operat????).mp,kw.	25
44	(ahead adj3 surgery).mp,kw.	52
45	(ahead adj3 surgical*).mp,kw.	13
46	(ahead adj3 procedur*).mp,kw.	13
47	(in advance adj3 surger*).mp,kw.	69
48	(in advance adj3 surgical*).mp,kw.	27
49	(in advance adj3 operat*).mp,kw.	40
50	(in advance adj3 procedur*).mp,kw.	50
51	(preceeding adj3 surger*).mp,kw.	7
52	(preceeding adj3 surg*).mp,kw.	11
53	(preceeding adj3 operat*).mp,kw.	6

54	(preceeding adj3 procedur*).mp,kw.	6
55	(preced* adj3 surger*).mp,kw.	795
56	(preced* adj3 surgical*).mp,kw.	287
57	(preced* adj3 operat*).mp,kw.	389
58	(preced* adj3 procedur*).mp,kw.	488
59	(prepar* adj1 surgery).mp,kw.	176
60	(surgery adj1 clinic?).mp,kw.	1430
61	(surgical adj1 clinic?).mp,kw.	2057
62	or/1-61 [Preoperative & related terms]	536329
63	Symptom Assessment/	5996
64	Data Collection/	90364
65	Diagnostic Tests, Routine/	13281
66	"Referral and Consultation"/	69151
67	"Surveys and Questionnaires"/	494935
68	Geriatric Assessment/	29161
69	Health Status Indicators/	23802
70	Interviews As Topic/	65116
71	Mass Screening/	107631
72	Narration/	8814
73	Patient Acuity/	1366
74	Patient Reported Outcome Measures/	8350
75	Patient Selection/	67183
76	exp Psychiatric Status Rating Scales/	85338
77	Psychological Tests/ or Trail Making Test/	37757
78	Self-report/	36360
79	Surgical Clearance/	37
80	(battery or batteries).mp,kw.	47381
81	(exam? or examin?? or examining or examination?).mp,kw.	2682278
82	(health adj2 determinant*).mp.	11645
83	(health adj2 determinant*).mp,kw.	11650
84	(routine?? adj1 (test or tests or tested or testing)).mp.	18165
85	(scale or scales).mp,kw.	741460
86	(score or scores or scored or scoring?).mp,kw.	894769
87	(patient? adj2 select*).mp,kw.	164908
88	(clear* adj3 (surgery or surgical*)).mp.	2990
89	(tool? or toolkit? or tool-kit?).mp,kw.	633651
90	assess*.mp,kw.	3054097
91	checklist?.mp,kw.	36824

92	check-list?.mp,kw.	2787
93	consultation?.mp,kw.	129947
94	evaluat*.mp,kw.	3546482
95	index*.mp,kw.	1033360
96	indicator?.mp,kw.	341259
97	indicator?.mp.	341259
98	indices.mp,kw.	154132
99	instrument?.mp,kw.	192834
100	interview*.mp,kw.	360296
101	inventories.mp,kw.	5929
102	inventory.mp,kw.	92926
103	measur*.mp,kw.	3227739
104	metric?.mp,kw.	51455
105	patient analys#s.mp,kw.	773
106	questionnaire?.mp,kw.	713256
107	rating?.mp,kw.	207450
108	screening?.mp,kw.	565121
109	selfassess*.mp,kw.	19
110	self-assess*.mp,kw.	24462
111	self-report*.mp,kw.	154670
112	status exam*.mp,kw.	1804
113	survey*.mp,kw.	975052
114	work-up?.mp,kw.	20217
115	workup?.mp,kw.	19473
116	or/63-115 [Assessment]	10890422
117	62 and 116 [Preoperative + Assessment]	354702
118	"Cell Phone Use"/	269
119	Cell Phones/	9046
120	exp Computer Communication Networks/	98493
121	exp Electronic Mail/	2775
122	exp Internet/	85053
123	Internet Access/ [MeSH 2019]	101
124	Internet-Based Intervention/ [New MeSH 2020]	553
125	"Internet of Things"/ [New MeSH 2020]	182
126	Mobile Applications/	7857
127	Online systems/	8439
128	Patient Portals/ [MeSH 2017]	508
129	Remote Consultation/	5184

130	Smartphone/	5957
131	Text Messaging/	3465
132	*Telephone/	5487
133	User-Computer Interface/	38147
134	"e-consult*".mp,kw.	162
135	"e-health*".mp,kw.	2653
136	"e-patient*".mp.	576
137	"health 2.0".mp.	106
138	"internet of things".mp.	868
139	"m-health*".mp,kw.	514
140	"ning.com".mp.	0
141	"second life".mp.	223
142	"u-health*".mp,kw.	43
143	"web 2.0".mp.	561
144	"www.ning.com".mp.	0
145	"www.patientslikeme.com".mp.	2
146	("on-line" adj6 (assess* or care or clinic? or consult* or evaluation? or planning or preplan*4 or treatment? or visit*)).mp,kw.	1162
147	(android adj2 (app or apps or application?)).mp.	382
148	(apple adj2 (app or apps or application?)).mp.	194
149	(communicat* adj10 platform?).mp.	952
150	(digital adj1 health*).mp,kw.	1887
151	(digital* adj10 platform?).mp.	1165
152	(digital?? adj6 outlet?).mp.	12
153	(digital?? adj6 portal?).mp.	179
154	(electronic* adj2 consult*).mp,kw.	303
155	(health care app or health care apps).mp.	21
156	(health* app or health* apps).mp.	609
157	(information adj2 highway*).mp.	127
158	(internet adj2 support*).mp.	433
159	(irc and relay).mp.	5
160	(mobile adj1 health*).mp,kw.	7364
161	(online adj2 account???).mp.	48
162	(on-line adj2 account???).mp.	7
163	(online adj2 support*).mp.	1179
164	(on-line adj2 support*).mp.	98
165	(online adj6 (assess* or care or clinic? or consult* or evaluation? or planning or preplan*4 or treatment? or visit*)).mp,kw.	8747

166	(portable adj2 (app or apps or application?)).mp.	182
167	(remote?? adj2 assess*).mp,kw.	511
168	(remote?? adj2 consult*).mp,kw.	5529
169	(remote?? adj2 evaluat*).mp,kw.	326
170	(short adj messag* adj1 service?).mp,kw.	946
171	(smart adj2 (phone? or device? or app or apps or application? or sensor?)).mp.	2111
172	(SMS and message?).mp,kw.	1121
173	(sociable adj2 technolog*).mp.	1
174	(social adj2 site?).mp.	1476
175	(social adj2 software*).mp.	226
176	(social adj2 technolog*).mp.	943
177	(social adj2 utilit*).mp.	239
178	(text? adj2 messag*).mp,kw.	5003
179	(video* adj2 assess*).mp,kw.	1762
180	(video* adj3 visit*).mp,kw.	434
181	(virtual adj2 collabor*).mp.	115
182	(virtual adj2 communit*).mp.	341
183	(virtual adj2 group*).mp.	327
184	(virtual adj2 support*).mp.	213
185	(virtual adj6 (assess* or care or clinic? or consult* or evaluation? or planning or preplan*4 or treatment? or visit*)).mp,kw.	5818
186	(web adj2 app?).mp.	249
187	(web adj2 application?).mp.	3028
188	(web adj2 support*).mp.	367
189	(webbased adj2 application?).mp.	1
190	(web-based adj2 application?).mp.	1078
191	bebo.mp.	22
192	cyber*.mp.	7416
193	digital tablet?.mp.	52
194	ecommunit*.mp.	5
195	e-communit*.mp.	31
196	econsult*.mp,kw.	112
197	ehealth*.mp,kw,jw.	3288
198	electronic mail*.mp.	3481
199	electronic support*.mp.	85
200	electronic tablet?.mp.	112
201	email*.mp.	6925
202	e-mail*.mp.	7365

203	epatient*.mp.	6
204	FaceTime??.mp.	60
205	google android.mp.	16
206	Google Group?.mp.	8
207	Google Hangout?.mp.	16
208	google pixel?.mp.	2
209	google play.mp.	252
210	Google Plus.mp.	9
211	Google Wave.mp.	4
212	GooglePlus.mp.	1
213	health* application?.mp.	1781
214	health* tech* enable?.mp.	11
215	instant messag*.mp.	284
216	inter* relay*.mp.	82
217	internet group*.mp.	104
218	internet* tablet?.mp.	10
219	internet*.mp,kw,jw.	100937
220	Kaixin001.mp.	0
221	mashup*.mp.	33
222	meebo.mp.	2
223	mhealth*.mp,kw,jw.	4213
224	Microsoft Surface??.mp.	2
225	mixi.mp.	3
226	myspace*.mp.	83
227	ning.mp.	243
228	orkut.mp.	2
229	personali#ed health*.mp.	829
230	phone app?.mp.	330
231	phone application?.mp.	467
232	secondlife.mp.	5
233	semantic web?.mp.	682
234	Sina Weibo.mp.	63
235	SinaWeibo??.mp.	1
236	Skype?.mp.	329
237	SnapChat??.mp.	66
238	tablet? computer?.mp.	658
239	Taringa?.mp.	1
240	telean?esth*.mp,kw.	9

241	tele-an?esth*.mp,kw.	5
242	teleassess*.mp,kw.	10
243	tele-assess*.mp,kw.	14
244	teleassist*.mp,kw.	29
245	tele-assist*.mp,kw.	43
246	telecare.mp,kw.	677
247	tele-care.mp,kw.	30
248	teleconsult*.mp,kw.	1254
249	tele-consult*.mp,kw.	96
250	telediagnos*.mp,kw.	189
251	tele-diagnos*.mp,kw.	32
252	telehealth*.mp,kw,jw.	5356
253	tele-health*.mp,kw,jw.	163
254	telematic?.mp,kw.	631
255	telemedic*.mp,kw.	31832
256	tele-medic*.mp,kw.	148
257	telesurg*.mp,kw.	288
258	tele-surg*.mp,kw.	34
259	text-messag*.mp,kw.	4944
260	uhealth*.mp,kw.	7
261	usenet*.mp.	24
262	videoassess*.mp,kw.	0
263	video-assess*.mp,kw.	246
264	videoconfer*.mp,kw.	3080
265	video-confer*.mp,kw.	744
266	video-visit*.mp,kw.	131
267	virtual communit*.mp.	291
268	virtual support*.mp.	48
269	web bas*.mp.	26993
270	web page*.mp.	1610
271	web site*.mp.	7088
272	webbas*.mp.	84
273	webcam*.mp.	316
274	web-cam*.mp.	143
275	weblog*.mp.	111
276	web-log*.mp.	58
277	webpage*.mp.	649
278	website*.mp.	22927

279	WeChat??.mp.	314
280	WhatsApp??.mp.	421
281	world wide web*.mp.	2867
282	www.mp.	1378
283	youtube*.mp.	1530
284	Zoom??.mp.	1516
285	or/118-284 [Virtual or Telemedicine or Telehealth]	280224
286	117 and 285 [Preoperative + Assessment + Virtual]	3849
287	limit 286 to english language	3567
288	limit 287 to "humans only (removes records about animals)"	3531
289	limit 288 to ("all infant (birth to 23 months)" or "all child (0 to 18 years)" or "newborn infant (birth to 1 month)" or "infant (1 to 23 months)" or "preschool child (2 to 5 years)" or "child (6 to 12 years)" or "adolescent (13 to 18 years)")	552
290	288 not 289	2979
291	288 not (adolescence or adolescent or adolescents or babies or baby or boy or boys or child or childhood or children or childrens or children's or fetus or fetal or foetus or foetal or girl or girls or infancy or infant or infants or neonatal or neonatally or neonate or neonates or newborn or newborns or paediatric or paediatrician or paediatricians or paediatrics or pediatric or pediatrician or pediatricians or pediatrics or preschool* or teen or teenage or teenagers or teens or toddler or toddlers or tween* or youth or youths).ti,jw.	3296
292	288 and (elder* or senior? or aged or adult* or man or men or woman or women).ti,jw.	92
293	limit 288 to ("all adult (19 plus years)" or "young adult (19 to 24 years)" or "adult (19 to 44 years)" or "young adult and adult (19-24 and 19-44)" or "middle age (45 to 64 years)" or "middle aged (45 plus years)" or "all aged (65 and over)" or "aged (80 and over)")	1651
294	or/290-293 [Limiting to Adult]	3353
295	remove duplicates from 294	3349

Table A3. Excluded studies and reasons for exclusion.

Study ID	Reasons for Exclusion
Vitkun (1995) [1]	Wrong study design
Vitkun (1999) [2]	Wrong study design
Basu (2001) [3]	Wrong study design
Lee (2003) [4]	Wrong outcome
VanDenKerkhof (2005) [5]	Wrong study design
Grant (2012) [6]	Wrong outcome
Dilisio (2014) [7]	Wrong study design
Latifi (2014) [8]	Wrong outcome

Ludbrook (2015) [9]	Wrong outcome
Zuidema (2015) [10]	Wrong outcome
Howell (2017) [11]	Wrong outcome
Mang (2019) [12]	Wrong outcome
Lu (2020) [13]	Wrong study design
Osman (2020) [14]	Wrong study design
Osman (2020) [15]	Wrong study design

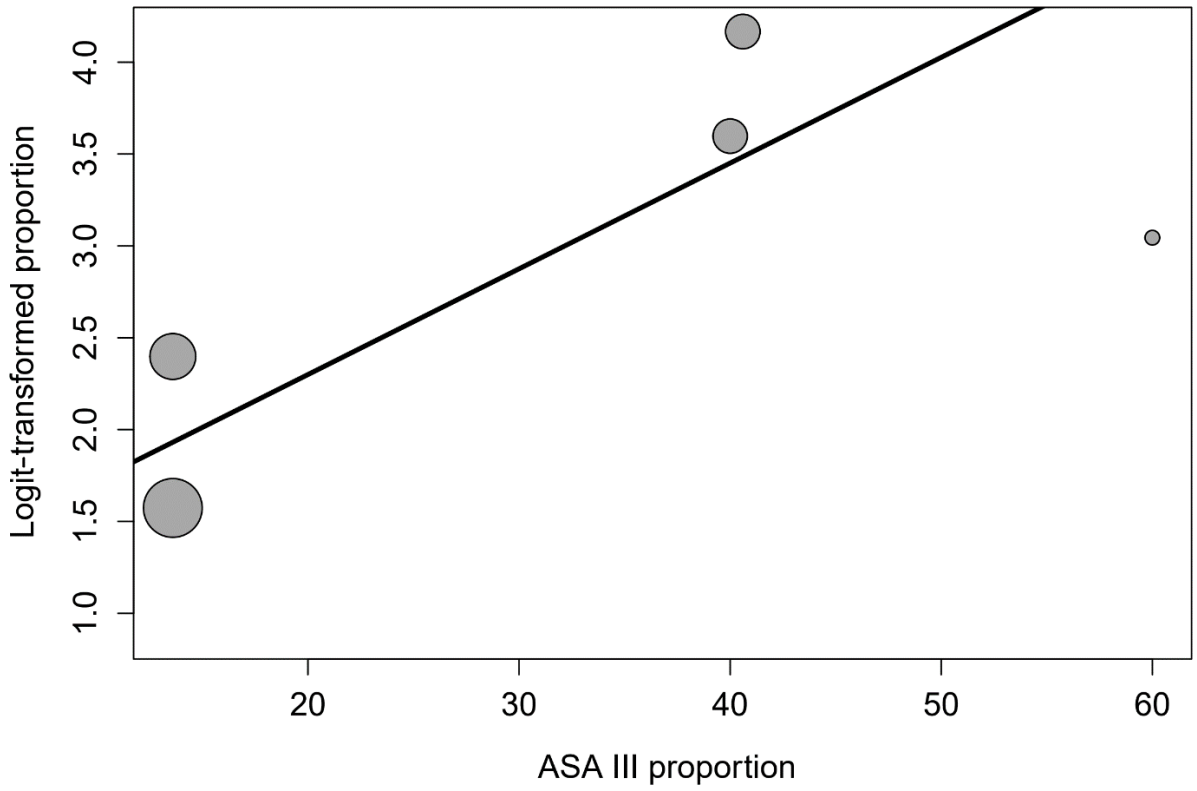


Figure A1. Bubble plot.

Bubble plot shows relationship between ASA III and positive patient experience estimates in the meta-regression analysis ($p < 0.01$).

Table A4. Newcastle-Ottawa Scale (NOS) – for cohort study.

Study	Selection				Comparability	Outcome			Total Score
	Representativeness of the exposed cohort (1)	Selection of the non-exposed cohort (1)	Ascertainment of exposure (1)	Demonstration that outcome of interest was not present at start of study (1)	Comparability of cohorts on the basis of the design or analysis (2)	Assessment of outcome (1)	Was follow-up long enough for outcomes to occur (1)	Adequacy of follow-up of cohorts (1)	
Studies with a comparison group									
Blanco Vargas (2012) [16]	*	*	*	*		*	*	*	7
Lozada (2016) [17]	*	*	*	*		*	*	*	7
Wood (2016) [18]	*		*	*		*	*	*	6
Goodhart (2017) [19]	*		*	*	*	*	*	*	7
Tam (2017) [20]	*		*	*		*	*	*	6
Afable (2018) [21]	*	*	*	*	*	*	*	*	8
Mullen-Fortino (2019) [22]	*	*	*	*	*	*	*		7
Kamdar (2020) [23]	*	*	*	*	**	*	*		8
O'Shea (2020) [24]	*	*	*	*	*	*	*		7
Studies without a comparison group									
Rollert (1999) [25]	*		*	*		*	*	*	6
Law (2009) [26]			*	*		*	*	*	5
Zetterman (2011) [27]	*		*	*		*	*	*	6

Table A5. Newcastle-Ottawa Scale (NOS) – for cross-sectional study.

Study	Selection				Comparability	Outcome		Total Score
	Representativeness of the sample (1)	**Sample size (1)**	**Non-respondents (1)**	**Ascertainment of exposure (2)**	**Comparability of subjects on the basis of the design or analysis (2)**	**Assessment of outcome (2)**	**Statistical test (1)**	
Roberts (2015) [28]	*			*		*		3

Table A6. JBI Critical Appraisal Checklist for Case Series.

Author: Wong et al. (2004) [29]

	Yes	No	Unclear	N/A
• Were there clear criteria for inclusion in the case series?	✓	☐	☐	☐
• Was the condition measured in a standard, reliable way for all participants included in the case series?	✓	☐	☐	☐
• Were valid methods used for identification of the condition for all participants included in the case series?	✓	☐	☐	☐
• Did the case series have consecutive inclusion of participants?	✓	☐	☐	☐
• Did the case series have complete inclusion of participants?	✓	☐	☐	☐
• Was there clear reporting of the demographics of the participants in the study?	✓	☐	☐	☐
• Was there clear reporting of clinical information of the participants?	☐	✓	☐	☐
• Were the outcomes or follow up results of cases clearly reported?	✓	☐	☐	☐

• Was there clear reporting of the presenting site(s)/clinic(s) demographic information?	☐	✓	☐	☐
• Was statistical analysis appropriate?	✓	☐	☐	☐

Overall appraisal: Include ✓ Exclude ☐ Seek further info ☐

Table A7. JBI Critical Appraisal Checklist for Randomized Controlled Trials.

Author: Applegate et al. (2013) [30]

	Yes	No	Unclear	N/A
1. Was true randomization used for assignment of participants to treatment groups?	✓	☐	☐	☐
2. Was allocation to treatment groups concealed?	✓	☐	☐	☐
3. Were treatment groups similar at the baseline?	✓	☐	☐	☐
4. Were participants blind to treatment assignment?	✓	☐	☐	☐
5. Were those delivering treatment blind to treatment assignment?	☐	✓	☐	☐
6. Were outcomes assessors blind to treatment assignment?	☐	☐	✓	☐
7. Were treatment groups treated identically other than the intervention of interest?	✓	☐	☐	☐
8. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and	✓	☐	☐	☐
9. Were participants analyzed in the groups to which they were randomized?	✓	☐	☐	☐
10. Were outcomes measured in the same way for treatment groups?	✓	☐	☐	☐
11. Were outcomes measured in a reliable way?	✓	☐	☐	☐
12. Was appropriate statistical analysis used?	✓	☐	☐	☐

13. Was the trial design appropriate, and any deviations from the standard RCT design (individual randomization, parallel groups) accounted for in the conduct and analysis of the trial? ✓ ☐ ☐ ☐

Overall appraisal: Include ✓ Exclude ☐ Seek further info ☐

References

[1] Vitkun SA, Gage JS, Anderson DH, Williams SA, Halpern-Lewis JG, Poppers PJ. Computerization of the preoperative anesthesia interview. Int J Clin Monit Comput 1995;12:71. https://doi.org/10.1007/BF01142486.

[2] Vitkun SA, Halpern-Lewis JG, Williams SA, Gage JS, Poppers PJ. Patient's perceptions of an anesthesia preoperative computerized patient interview. J Clin Monit Comput 1999;15:503–7. https://doi.org/10.1023/A:1009944409565.

[3] Basu S, Babajee P, Selvachandran SN, Cade D. Impact of questionnaires and telephone screening on attendance for ambulatory surgery. Ann R Coll Surg Engl 2001;83:329–31.

[4] Lee S, Broderick TJ, Haynes J, Bagwell C, Doarn CR, Merrell RC. The role of low-bandwidth telemedicine in surgical prescreening. J Pediatr Surg 2003;38:1281–3. https://doi.org/10.1016/S0022-3468(03)00382-8.

[5] VanDenKerkhof EG, Goldstein DH, Blaine WC, Rimmer MJ. A Comparison of Paper with Electronic Patient-Completed Questionnaires in a Preoperative Clinic. Anesth Analg 2005;101:1075–80. https://doi.org/10.1213/01.ane.0000168449.32159.7b.

[6] Grant C, Ludbrook GL, O'Loughlin EJ, Corcoran TB. An Analysis of Computer-Assisted Pre-Screening Prior to Elective Surgery. Anaesth Intensive Care 2012;40:297–304. https://doi.org/10.1177/0310057X1204000213.

[7] Dilisio RP, Dilisio AJ, Weiner MM. Preoperative virtual screening examination of the airway. J Clin Anesth 2014;26:315–7. https://doi.org/10.1016/j.jclinane.2013.12.010.

[8] Latifi R, Mora F, Bekteshi F, Rivera R. Preoperative telemedicine evaluation of surgical mission patients: should we use it routinely? Bull Am Coll Surg 2014;99:17–23.

[9] Ludbrook G, Seglenieks R, Osborn S, Grant C. A call centre and extended checklist for pre-screening elective surgical patients – a pilot study. BMC Anesthesiol 2015;15:77. https://doi.org/10.1186/s12871-015-0057-1.

[10] Zuidema X, Leuverink T, Houweling P. Validation of a Patient Self-Administered Pre-Anaesthetic Screening Questionnaire. Ambul Surg 2015;21:181–5.

[11] Howell M, Hood AJ, Jayne DG. Use of a patient completed iPad questionnaire to improve pre-operative assessment. J Clin Monit Comput 2017;31:221–5. https://doi.org/10.1007/s10877-015-9818-0.

[12] Mang J, Schild S, Prokosch H-U, Jeleazcov C, Heinrich A, Toddenroth D. Evaluation of a Patient-Facing Digital Prototype for Perioperative Risk Assessment. Stud Health Technol Inform 2019;258:201–5.

[13] Lu AC, Schmiesing CA, Mahoney M, Cianfichi L, Semple AK, Watt D, et al. COVID-19 Preoperative Assessment and Testing: From Surge to Recovery. Ann Surg 2020;272:e230–5. https://doi.org/10.1097/SLA.0000000000004124.

[14] Osman T, Lew E, Lum E, Chew J, Dabas R, Sng BL, et al. Effect of PreAnaesThesia Computerized Health (PATCH) Assessment on Duration of Nurse—Patient Consultation and Patient Experience: A Pilot Trial. Int J Environ Res Public Health 2020;17:4972. https://doi.org/10.3390/ijerph17144972.

[15] Osman T, Lew E, Lum EP-M, van Galen L, Dabas R, Sng BL, et al. PreAnaesThesia computerized health (PATCH) assessment: development and validation. BMC Anesthesiol 2020;20:286. https://doi.org/10.1186/s12871-020-01202-8.

[16] Blanco Vargas D, Faura Messa A, Izquierdo Tugas E, Santa-Olalla Bergua M, Noguera Sopeña MM, Mañoso Noriego M. Evaluación preoperatoria «on line» frente a evaluación presencial: optimización de recursos. Rev Esp Anestesiol Reanim 2012;59:350–6. https://doi.org/10.1016/j.redar.2012.05.039.

[17] Lozada MJ, Nguyen JTC, Abouleish A, Prough D, Przkora R. Patient preference for the pre-anesthesia evaluation: Telephone versus in-office assessment. J Clin Anesth 2016;31:145–8. https://doi.org/10.1016/j.jclinane.2015.12.040.

[18] Wood EW, Strauss RA, Janus C, Carrico CK. Telemedicine Consultations in Oral and Maxillofacial Surgery: A Follow-Up Study. J Oral Maxillofac Surg 2016;74:262–8. https://doi.org/10.1016/j.joms.2015.09.026.

[19] Goodhart IM, Andrzejowski JC, Jones GL, Berthoud M, Dennis A, Mills GH, et al. Patient-completed, preoperative web-based anaesthetic assessment questionnaire (electronic Personal Assessment Questionnaire PreOperative): Development and validation. Eur J Anaesthesiol 2017;34:221–8. https://doi.org/10.1097/EJA.0000000000000545.

[20] Tam A, Leung A, O'Callaghan C, Fagermo N. Role of telehealth in perioperative medicine for regional and rural patients in Queensland: Telehealth in perioperative medicine. Intern Med J 2017;47:933–7. https://doi.org/10.1111/imj.13484.

[21] Afable MK, Gupte G, Simon SR, Shanahan J, Vimalananda V, Kim EJ, et al. Innovative Use Of Electronic Consultations In Preoperative Anesthesiology Evaluation At VA Medical Centers In New England. Health Aff 2018;37:275–82. https://doi.org/10.1377/hlthaff.2017.1087.

[22] Mullen-Fortino M, Rising KL, Duckworth J, Gwynn V, Sites FD, Hollander JE. Presurgical Assessment Using Telemedicine Technology: Impact on Efficiency, Effectiveness, and Patient Experience of Care. Telemed J E Health 2019;25:137–42. https://doi.org/10.1089/tmj.2017.0133.

[23] Kamdar NV, Huverserian A, Jalilian L, Thi W, Duval V, Beck L, et al. Development, Implementation, and Evaluation of a Telemedicine Preoperative Evaluation Initiative at a Major Academic Medical Center. Anesth Analg 2020;131:1647–56. https://doi.org/10.1213/ANE.0000000000005208.

[24] O'Shea J, Coleman M, Mahdy S, Corbett M, Curley G. Evaluation of an online preoperative assessment tool. BMJ Innov 2020;6:127–31. https://doi.org/10.1136/bmjinnov-2019-000344.

[25] Rollert MK, Strauss RA, Abubaker AO, Hampton C. Telemedicine consultations in oral and maxillofacial surgery. J Oral Maxillofac Surg 1999;57:136–8. https://doi.org/10.1016/S0278-2391(99)90226-4.

[26] Law TT, Suen DTK, Tam YF, Cho SY, Chung HP, Kwong A, et al. Telephone pre-anaesthesia assessment for ambulatory breast surgery. Hong Kong Med J 2009;15:179–82.

[27] Zetterman CV, Sweitzer BJ, Webb B, Barak-Bernhagen MA, Boedeker BH. Validation of a virtual preoperative evaluation clinic: a pilot study. Stud Health Technol Inform 2011;163:737–9.

[28] Roberts S, Spain B, Hicks C, London J, Tay S. Telemedicine in the Northern Territory: An assessment of patient perceptions in the preoperative anaesthetic clinic: Telemedicine in the Northern Territory. Aust J Rural Health 2015;23:136–41. https://doi.org/10.1111/ajr.12140.

[29] Wong DT, Kamming D, Salenieks ME, Go K, Kohm C, Chung F. Preadmission Anesthesia Consultation Using Telemedicine Technology: A Pilot Study. Anesthesiology 2004;100:1605–7. https://doi.org/10.1097/00000542-200406000-00038.

[30] Applegate RL, Gildea B, Patchin R, Rook JL, Wolford B, Nyirady J, et al. Telemedicine Pre-anesthesia Evaluation: A Randomized Pilot Trial. Telemed J E Health 2013;19:211–6. https://doi.org/10.1089/tmj.2012.0132.